CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (All Branches)

May 2014

CL103/CL103.01 Mechanics of Solids

Date: 24.05.2014, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Find centroid of the shaded area shown in Figure 1. **[07]**

Q - 2 (a) The force F1 is of 100N magnitude. The resultant of the force F1 and unknown force F2 is 100N in magnitude. Find the magnitude of F2 if resultant is perpendicular to F1. **[04]**

(b) Attempt any Two. **[10]**

(i) A 50kN is the resultant of four forces out of which three are shown in Figure 2. Find thee direction and magnitude of the fourth force.

(ii) Two cylinders, each of diameter 40cm and mass 500N, rest in a rectangle channel of base width 72cm. Find the reactions at all contact surfaces. Refer Figure 3.

(iii) Determine the resultant of the four forces shown in Figure 4 with respect to point B.

Q - 3 (a) A uniform ladder of length 4m placed against a vertical wall making an angle of 45^0 with the floor. The coefficient of friction between the wall and the ladder is 0.4 and that between floor and ladder is 0.5. If a man whose weight is ½ of that of ladder ascends it, at what height will he be when the ladder slips? **[07]**

(b) Determine forces in all members of any one truss. Refer Figure 5 and Figure 6. **[07]**

SECTION – II

Q - 4 Define stiffness, toughness and brittleness. Explain stress-strain curve of a ductile material when subjected to a tensile loading up to failure. **[07]**

Q - 5 (a) Derive $\Delta l = 4PL/\pi E d_1 d_2$ with usual notations. **[04]**

OR

Q - 5 (a) Explain SDOF & MDOF systems with suitable examples. **[04]**

(b) Attempt any Two. **[10]**

(i) A steel rail is 12m long and is laid at a temperature of 18^0C. The maximum temperature expected is 40^0C.

   a. Estimate the minimum gap between rails to be left so that the temperature stresses do not develop.
   b. Calculate the temperature stresses developed in the rails if:
      1. No expansion joint is provided.
      2. If a 1.5 mm gap is provided for expansion

(ii) Find out force P required for equilibrium of the bar shown in Figure 7. Find out total deformation of the bar if Es = 200GPa, Eb = 100GPa, Eal = 75GPa.

(iii) A bar of brass 20mm is enclosed in a steel tube of 40mm external diameter and 20mm internal diameter. The bar and the tubes are initially 1.2 m long and are rigidly fastened at both ends using 20mm diameter pins. If the temperature is raised by 60^0C, find the stresses induced in the bar, tube and pins.

Q – 6 Draw shear force and bending moment diagrams for **any two beams** shown in Figure 8, 9, 10 indicating values at all important points. Locate point of contra flexure if any. [14]

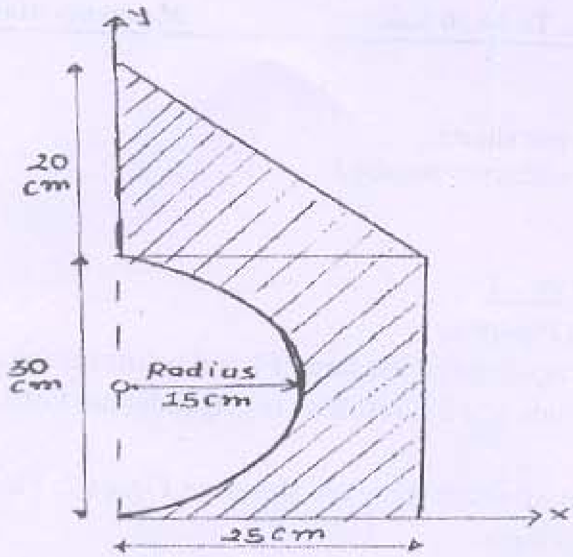

Figure 1

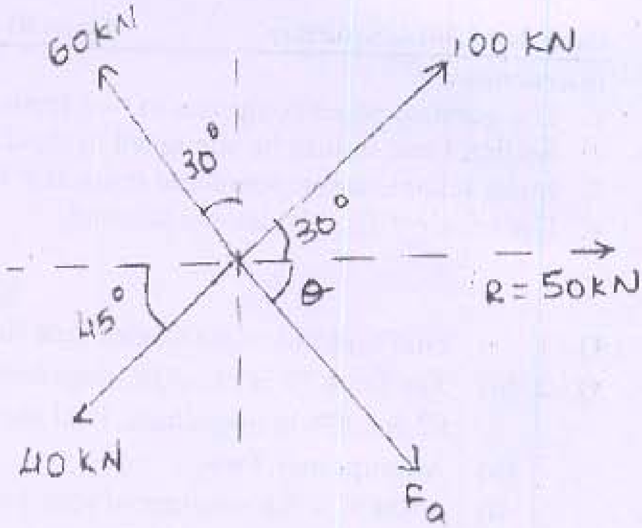

Figure 2

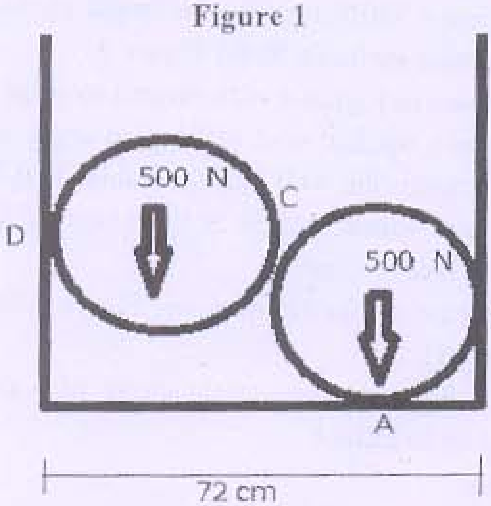

Figure 3

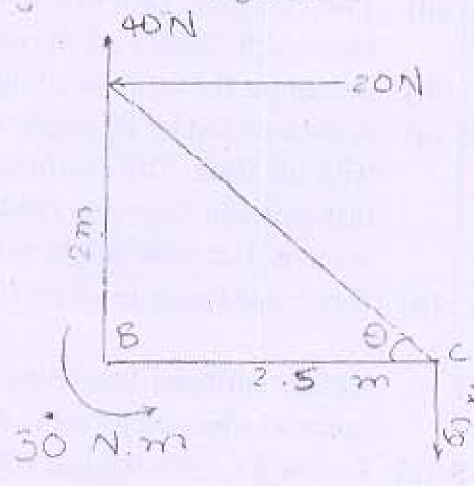

Figure 4

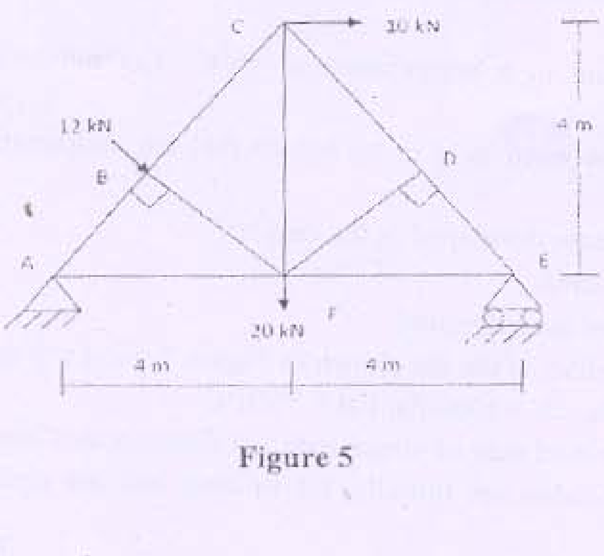

Figure 5

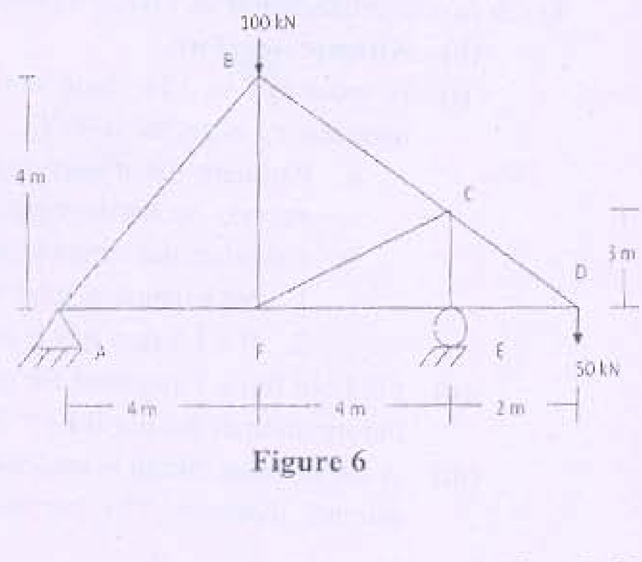

Figure 6

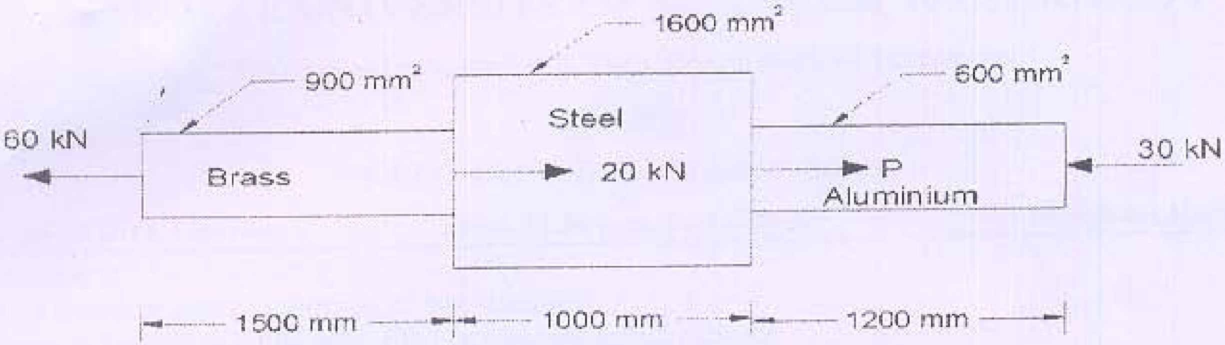

Figure 7

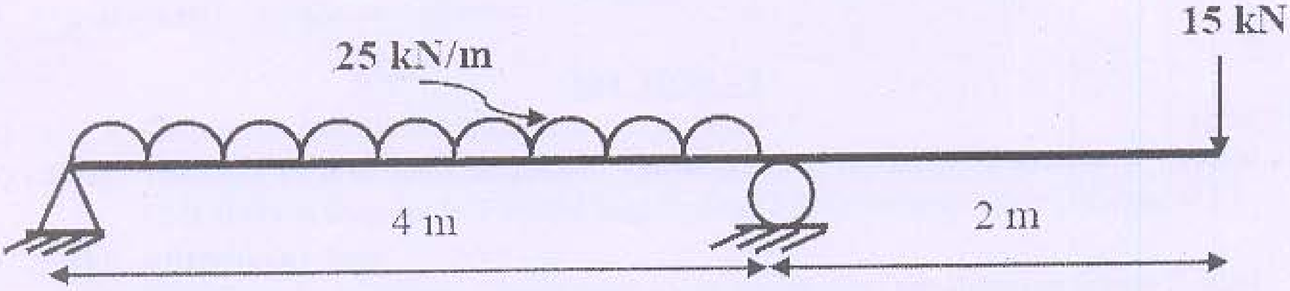

Figure 8

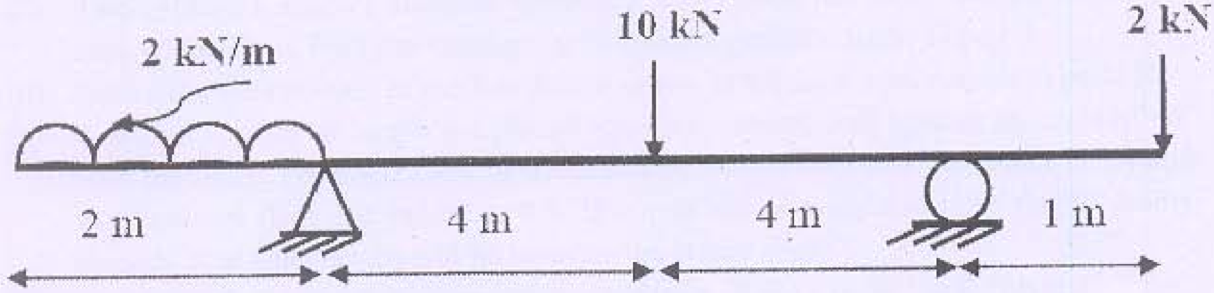

Figure 9

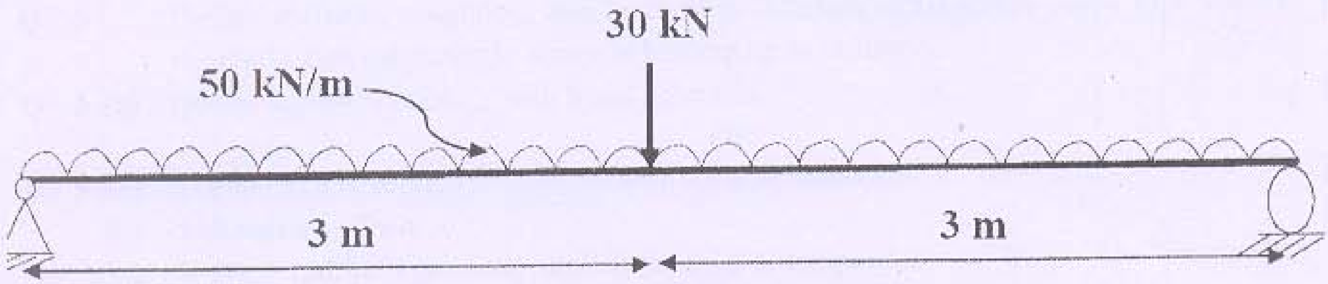

Figure 10

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